Lesson Notes

MODULE 6 | LESSON 3

K2: LEARNING BAYESIAN NETWORK STRUCTURE

Reading Time	30 minutes
Prior Knowledge	Bayesian network basics
Keywords	K2 algorithm

In the last lesson, we talked about structure learning at a high level and made specific reference to the Chow-Liu algorithm. In this lesson, we continue that discussion, focusing on the work of Cooper and Herskovits, especially their contribution to the field, K2.

1. A Classic in Bayesian Network Research

The main reading for this lesson is a classic, which implies that it is old but valuable. So let's get this out of the way: Since this paper is rather old relative to much of the research we have been reading in this course, some of the language feels a bit outdated. However, if you make a mental substitution of the words "dataset" whenever you read Cooper and Herskovits write "database," you will better appreciate the article's (early but important) contribution to this field. Similarly, when Cooper and Herskovits write about "cases" (309-10, etc.), we could read instead the word "observations" and their research would not feel out of step with today's data science vernacular. If we were speaking probabilistically, we could instead say "outcomes." Another note on terminology: Many researchers, including Cooper and Herskovits, use the term Bayesian belief networks to what we have been calling simply Bayesian network. That is their preferred term, but it is a distinction without a difference. Some authors use the terms interchangeably; some authors also clarify that they mean the same thing whichever term they use. Here are those updates in summary:

- database -> dataset
- case -> observation
- Bayesian belief network <-> Bayesian network

In this this lesson, we read some early work on Bayesian networks, giving us the techniques for identifying the dependence structure in the data that lends itself to the directed acyclic graph that will serve as our Bayesian network, much like the Chow-Liu algorithm.

The primary contribution from the required reading paper by Cooper and Herskovits is the K2 algorithm. They demonstrate the performance of this algorithm by testing whether it can identify the

ALARM network, which is well-known for capturing complex dependence relationships among variables. So we will discuss both the K2 and ALARM network below.

Cooper and Herskovits refer the reader to a 1988 work by Judea Pearl for “a detailed discussion of the semantics of Bayesian belief networks" (311), but hopefully, you find that the authors’ own discussion explains how many of the concepts from the previous lessons of this module come together logically. Indeed, you should see this lesson’s reading as the consolidation of what you know, along with a moderate amount of expansion on these ideas and techniques.

Since you have seen similar (but different) techniques employed towards similar (but different) ends, you will not be surprised to read that Cooper and Herskovits use “Monte Carlo simulation” (310-1) to generate the observations on which they base their probability assignments in their table 2. This differs (but not too much) from the way that you used the `sample` method (which randomly sampled from the conditional probabilities tables, or CPT, provided) from Halford’s Sorobn repo, and then determined a structure based on those samples with the `chow_liu` function. You might even say that the Cooper and Herskovits approach here is more direct: They sample directly from the observations of the dataset, while in this scenario, Halford samples from the CPTs, which presumably, though not necessarily, are themselves based on a data set (consisting of observations). This is not, in fact, meant to contrast Halford’s and Cooper-Herskovits's overall approaches. This is only meant to orient aspects of the current reading in the context of concepts and operations you have already seen.

The other reason this paper is valuable to us for our purposes is the “Introduction” section itself (309-13), which formalizes much of the theoretical material we have covered so far, while also providing insightful background about the practical challenges in the field of Bayesian networks, such as the time-consuming reliance on experts (312).

As you work through the paper, you will note the nifty observation that the posterior probability can often require a cumbersome calculation, but that the ratio of two network’s posterior probabilities is equal to the ratio of their joint probabilities, which is much more straightforward and tractable, especially given a small set of initial assumptions:

1. The variables in our dataset are discrete, which we assumed with the Sorobn repo as well: True/False, Good/Bad, High/Low, etc.
2. Observations are independent, for a given network.
3. There is no missing data. (This assumption is relaxed later in the paper).
4. The probability density function (pdf) of the conditional probabilities that make up a network is the well-known and easy-to-work-with Uniform distribution. To make this more clear, once we have assumed a network structure, we do not impose any prior belief on the conditional probabilities, implying the following:

$$P(X = x|Y = y) \sim Uniform(0, 1)$$

Note 1: We are implicitly referring to Bayesian networks as a composition of a network structure and conditional probabilities. If we have the same structure, but the conditional probabilities are different in any way, then we have two distinct Bayesian networks; by the same token, perhaps more obviously, if the structures are in any way different, we again have two distinct Bayesian networks (though in that case, the conditional probabilities would likely also differ in at least some respects).

Note 2: We say pdf here because we are referring to the continuous variable (on the support [0,1]) that represents the probability value. The probabilities themselves still refer to discrete values for the variables. This in turn leads to the use of multiple integrals to find the joint probabilities of the Bayesian networks and the data.

Note 3: This assumption also gets relaxed later in the paper so that the relevant distribution is the more general Dirichlet distribution, itself the conjugate prior for the Categorical distribution. Given our assumed restriction to discrete variables (first assumption), the Categorical distribution should seem like a natural and reasonable direction for the relaxation of the fourth assumption.

2. K2, or Why Do We Need an Algorithm to Find the Optimal Network?

After reading the first part of “Section 3.1: Finding the most probably belief-network structures” and admiring the recursive function for determining the possible number of structures for a given number of nodes n, take note of its results and be astounded. This super-exponential growth in the number of possible structures is why a heuristic method for identifying strong-candidate structures is needed:

- For $n = 2, there are$three possible structures.
- For $n = 3, there are$twenty-five possible structures.
- For $n = 5$, there are 29,000 possible structures.
- For $n = 10$, _there are _4.2×10^18 possible structures.

We obviously cannot be in the business of searching such numbers of structures exhaustively, especially if we have have $n > 10,$ which would be quite realistic. This is why we need K2, a greedy heuristic algorithm that “proceeds by first assuming a node has no parents and then incrementally adding as a parent that predecessor in the ordering whose addition most increases the probability of the resulting structure until no improvement is possible or a pre-specified threshold on the maximum number of parents is reached” (Thomas). The following pseudo-code is offered as a more readable version of the K2 algorithm that you find in the reading.

Input: Order of variables, data set, max_parents

Output: Network structure

For i = 1 to n:

$\pi_i = \emptyset$

$p_old = g(i, \pi_i)

ok_to_proceed = True

While (ok_to_proceed) and (| π_i | < max_parents):

$z \in pred(i) where z =$argmax_k g(i, \pi_i \cup {k})$

$p_new = g(i, \pi_i \cup k)$

If(p_new > p_old):

p_old = p_new

π_i =\pi_i = \cup {z})$

Else:

Ok_to_proceed = False

(Sierra et al. 6)

3. The Advantages of Orderings

You must also read the “Section 3.1.1: Exact methods,” but do not worry much about knowing the formulas and their derivations, at least not for the quiz for this lesson. Perhaps the most important technical concept from this section is the description of ordering constraint, where the ordering has not yet been well defined—though it could be as simple as (or involve) temporal ordering as in the case of the stock market opening and closing times across the globe that we saw in the last lesson.

Later researchers approvingly cite heuristic methods that are based on orderings, such as K2, for at least three reasons:

1. “If the search is conducted in the space of orderings, instead of networks structures, there is a severe decrease in the search space, which eases the task” (Carvalho 2).
2. “Given a network consistent with some ordering on the variables, there is no need to check for network cycles, since it is guaranteed that the network will always be acyclic” (Carvalho 2).
3. “The optimal network consistent with the topological order will always score better than, or the same as, the optimal branching” (Carvalho 2).

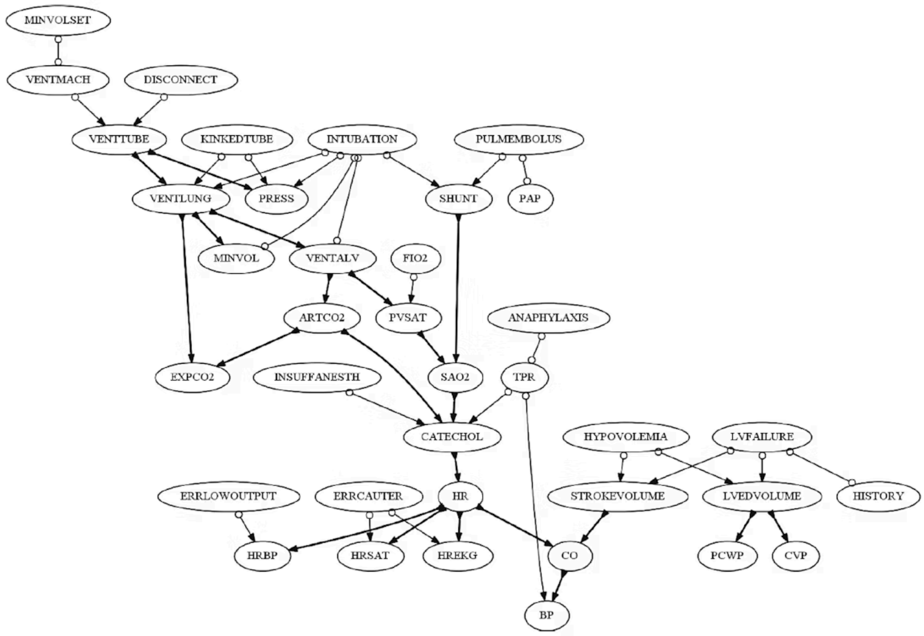
4. The ALARM Network

The authors provide a superficial description of the ALARM Network “as an initial research prototype to model potential anesthesia problems" (Cooper and Herskovits)—a risk management problem indeed, even if not a financial one. The creator of the network provides some additional background that elucidates its complexity and underlines its intrinsic connection to our focus:

“ALARM (A Logical Alarm Reduction Mechanism) is a diagnostic application used to explore probabilistic reasoning techniques in belief networks. ALARM implements an alarm message system for patient monitoring; it calculates probabilities for a differential diagnosis based on available evidence. The medical knowledge is encoded in a graphical structure connecting 8 diagnoses, 16 findings and 13 intermediate variables” (Beinlich).

Furthermore, while Cooper and Herskovits have removed the text descriptions of each variable in favor of natural numbers to avoid clutter, below we replace the short descriptions with the hope that the supplemental context may help you interpret and rationalize the structure (despite its complexity) and develop a better intuition about network structures generally.

Figure: ALARM Network with Text Labeled Nodes



Source: Isozaki, *Takashi,* and Manabu Kuroki. "Learning Maximal Ancestral Graphs with Robustness for Faithfulness Violations." **Advanced Methodologies for Bayesian Networks**. AMBN 2015, Lecture Notes in Computer Science, Springer, Cham. vol. 9505, Nov. 2015, https://doi.org/10.1007/978-3-319-28379-1_14.

Obviously, this use case has little to do directly with financial engineering, but it is meant to convey the complexity that the K2 algorithm can learn and that Bayesian networks can capture once the structure has been learned.

5. Conclusion

In this lesson, we discussed the classic K2 algorithm while the reading spelled out the K2 scoring metric so that in the next lesson we are prepared to work with the code for these and other closely related algorithms and measures.

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